

Week 02: VAE and GAN model ideas

Generative AI | MSc teaching notes and further reading | 2027

Learning target

Explain how reconstruction differs from adversarial generation.

Context and teaching plan

A designer wants new simple icons similar to examples, without copying an exact icon.

Contact time	Suggested use
Two hours: concepts	Case introduction, four ideas, worked example and misconception.
Two hours: practice	10 min recall, 20 min demo, 40 min guided practice, 25 min independent change, 15 min review, 10 min exit check.

Explain the key ideas

VAE

An encoder predicts the centre (μ) and spread (σ) of a small latent distribution. We sample a code z from it. A decoder uses z to reconstruct the image.

Teacher question: Where does vae appear in the weekly case, and what could go wrong if we omit it?

Latent space

Nearby latent values often produce related outputs, though this is learned rather than guaranteed.

Teacher question: Where does latent space appear in the weekly case, and what could go wrong if we omit it?

GAN

A generator proposes samples. A discriminator learns to distinguish generated and real samples; both change during training.

Teacher question: Where does gan appear in the weekly case, and what could go wrong if we omit it?

Training curves

A lower loss alone does not certify useful images; inspect quality, diversity and a held-out sample.

Teacher question: Where does training curves appear in the weekly case, and what could go wrong if we omit it?

Worked example and answer guide

- Input: Input icon represented by [0.2, 0.8].
- Change or question: Latent coordinate z changes from 0.3 to 0.7.
- Supported observation: Toy decoder output changes; inspect rather than assume improvement.

Reasoning: state the given evidence first, then the calculation or rule, then the conclusion. If a value is absent, say what cannot be concluded. Keep any simulated mechanism distinct from a trained model or deployed service.

Teach the week as a story

The icon studio: A designer wants varied icons that resemble a style without copying one example. How could two training ideas approach this?

Ask for a quick prediction before each technical explanation. Return to the prediction after the example and ask what evidence changed the answer. The analogies below are memory aids, and each has a stated limit.

VAE: analogy, mechanism and check

Familiar picture: A travel suitcase packs the essential features and unpacks them at the destination.

Familiar part	Technical meaning
Packing	encoder
Compact code	sampled latent z
Unpacking	decoder

Where this analogy breaks: A VAE learns a distribution and reconstruction objective; a suitcase is not a probabilistic model.

Actual idea: An encoder predicts the centre (μ) and spread (σ) of a small latent distribution. We sample a code z from it. A decoder uses z to reconstruct the image.

Work it aloud: A 2-value icon [0.2, 0.8] enters an encoder; a sampled z is decoded into a reconstructed icon.

Ask before revealing: Which arrow could lose a fine detail?

Supported response: Compression into z can discard information; compare original and reconstruction.

Look for this misconception: A VAE stores and copies exact training icons.

Latent space: analogy, mechanism and check

Familiar picture: A map places similar destinations near one another.

Familiar part	Technical meaning
Map point	latent coordinate
Nearby points	related codes
Visit	decoded output

Where this analogy breaks: Closeness in a toy plot does not guarantee human-perceived similarity.

Actual idea: Nearby latent values often produce related outputs, though this is learned rather than guaranteed.

Work it aloud: Change toy z from 0.3 to 0.7; compare the output pattern while keeping the decoder fixed.

Ask before revealing: Does a smoother output prove realism?

Supported response: No. Inspect examples and use a suitable quality test.

Look for this misconception: A latent coordinate is a pixel location.

GAN: analogy, mechanism and check

Familiar picture: A forger and examiner improve through repeated feedback.

Familiar part	Technical meaning
Forger	generator
Examiner	discriminator
Critique	training signal

Where this analogy breaks: The analogy can imply a human examiner; the discriminator is a learned network and may be fooled.

Actual idea: A generator proposes samples. A discriminator learns to distinguish generated and real samples; both change during training.

Work it aloud: A generator proposes an icon; the discriminator scores real versus generated examples; both update.

Ask before revealing: Which part produces the new image?

Supported response: The generator produces it; the discriminator provides a training signal.

Look for this misconception: The discriminator paints the final image.

Training curves: analogy, mechanism and check

Familiar picture: A student gets better on practice sheets while worse on unfamiliar exam sheets.

Familiar part	Technical meaning
Practice curve	training loss
Unseen sheet	validation set
Visual inspection	quality check

Where this analogy breaks: GAN losses are not simple universal quality scores; compare held-out samples too.

Actual idea: A lower loss alone does not certify useful images; inspect quality, diversity and a held-out sample.

Work it aloud: Training error $0.30 \rightarrow 0.18 \rightarrow 0.12$; validation $0.32 \rightarrow 0.20 \rightarrow 0.26$. Inspect epoch 3.

Ask before revealing: Why is the final training number insufficient?

Supported response: Validation worsened; quality and variety require separate inspection.

Look for this misconception: Lower training loss always proves better new icons.

New situation for discussion

The generated icon looks sharp but repeats one shape. Which additional check would reveal lack of diversity?

Memory cue: VAE: compress and reconstruct. GAN: propose and critique.

Guided laboratory

Annotate model paths and explore a toy latent coordinate.

1. Trace an example through encoder, sample and decoder
2. Trace random input through generator and discriminator
3. Inspect supplied loss and quality figures
4. Change one latent coordinate in a toy visualization

Individual change: Change z and compare output difference; explain that the toy cannot establish realism. Expected evidence: Two labelled model diagrams and a reconstruction comparison.

Checks and common error

Measure: Mean absolute reconstruction error on supplied example pairs. Record the input, setting, source version, result and the relevant numerator and denominator.

Plausible incorrect claim: The toy decoder is a trained VAE or a trained GAN.

Correction: It is only a deterministic illustration of latent variation. Real models learn parameters from examples.

If learners disagree, reproduce the same case and compare the underlying source or expected answer. Record unresolved cases for instructor review.

Further discussion

- How could the conclusion change if the source or input changes?
- Which part of the result is observed and which part is an assumption?
- What simple test would detect the failure described above?

References and optional extension

Core readings are official vendor documentation or original research. Read the relevant section; an account or paid API is unnecessary for this notebook. Page details may change before 2027.

Google, GAN overview. <https://cloud.google.com/discover/what-are-generative-adversarial-networks>

NVIDIA, NeMo framework overview.

<https://docs.nvidia.com/nemo-framework/user-guide/latest/overview.html>

Optional extension: replace one classroom proxy with a locally approved model or service, keep the original baseline, repeat the same checks, and state the additional privacy and cost assumptions.

Curriculum scope: the supplied 12-week teaching sequence, recorded in docs/SYLLABUS_ALIGNMENT.md. The original module descriptor is retained in sources/originals/07-Descriptor-Generative-AI-new-.docx. Supporting project notes: Deep-research.txt, RAG.txt, AI-Agents.txt, MCP-&-Vector-Database.txt, Multi-Agent-AI-System.txt, Eval-&-Finetune-&-Loop-Engineering.txt. These are planning sources, not empirical performance evidence.

Student revision and exam preparation

Outcomes: LO1, LO2 | 2 hours concepts + 2 hours guided practice

The idea in plain language

A VAE learns to describe an example using a latent distribution and reconstruct it. A GAN learns through a generator and a discriminator. In both cases inspect outputs as well as numerical loss. GAN losses depend on both networks, so a falling value alone does not prove improvement.

A worked example

Supplied VAE reconstruction errors fall from 0.30 to 0.18 to 0.12, while validation error falls to 0.20 then rises to 0.26. This can indicate overfitting. A GAN repeatedly producing the same icon may have poor diversity even when its images look sharp.

Practice before reading the answers

1. Label input, encoder, latent sample and decoder in a VAE. (4 suggested practice marks)

2. Training error falls but validation error rises. What should you inspect? (4 suggested practice marks)

3. A GAN produces only one excellent icon. Is it a successful generator? (4 suggested practice marks)

Draw the main diagram from memory, explain each arrow, then check it against the slides. Answer the three questions without notes before reading the model answers.

Model answers and marking guidance

1. Input enters encoder. Encoder predicts a latent distribution. Sample z from that distribution. Decoder reconstructs the input from z .

2. Inspect held-out examples and overfitting; compare checkpoints and avoid claiming progress from training error alone.

3. Not necessarily. Quality may be high but diversity is poor, consistent with mode collapse.

For each four-mark practice answer: two marks for the correct idea, one for a relevant example or calculation, one for an appropriate explanation or limitation. These suggested practice marks do not change the module assessment scheme.

Official reading and video support

G2 Google Developers: Overview of GAN Structure

https://developers.google.com/machine-learning/gan/gan_structure

Study focus: Generator, discriminator and training feedback.

G3 Google / TensorFlow: Convolutional Variational Autoencoder

<https://www.tensorflow.org/tutorials/generative/cvae>

Study focus: VAE architecture; core students interpret prepared examples rather than train it.

N2 NVIDIA: NeMo Framework overview

<https://docs.nvidia.com/nemo-framework/user-guide/latest/overview.html>

Study focus: Instructor background on model customization; no NVIDIA setup required in core labs.

Selected source concepts inform the lesson. Examples, figures, questions and solutions are original classroom material; no exam questions are claimed to be official.